

treatment planning (CT and MRI) and treatment delivery, patients were immobilized with a BrainLAB thermoplastic mask. Eye immobilization was achieved by instructing the patient to fixate a light source integrated into the mask system. A mini-video camera was used to provide on-line information about the eye and pupil position, respectively. A new CT and MR compatible prototype, based on head & neck fixation and the infrared tracking system ExacTrac has been developed and evaluated since 2002. This system records maximum temporal and angular deviations during treatment and based on tolerance limits, a feedback signal to the linac enables gated SRT.

Results: After a median follow-up of 33.4 months (range 3–85 months) local control was achieved in 98%. Fifteen patients (9.0%) developed metastases. Secondary enucleation was performed in 23 patients (13.8%). Long term side effects were retinopathy (n=70; 44%), cataract (n=30; 23%), optic neuropathy (n=65; 41%), and secondary neovascular glaucoma (n=23; 13.8 %). Typical situations when preset deviation criteria were exceeded were slow drifts (fatigue), sudden large eye movements (irritation), or if patients closed their eyes (fatigue). In these cases radiation was reliably interrupted by the gating system. In our clinical setup the novel system for computer-controlled gated SRT of uveal melanoma was well tolerated by about 30 patients treated so far with this system.

Conclusions: Linac based SRT of uveal melanomas provides good local control. Together with the possibility of performing movement-gated treatments, the new prototype system improves the quality of treatment. In an ongoing study treatment related side effects are correlated with dose levels. Such correlations can be used to further optimize linac based SRT of uveal melanoma.

1024 Visual Outcomes and Eye Preservation After Iodine-125 Brachytherapy for Uveal Melanoma

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Purpose/Objective: Since the publication of the landmark COMS trial, plaque radiotherapy has been established as a safe and effective alternative to enucleation for the treatment of uveal melanoma. However, the literature is largely silent regarding visual outcomes after Iodine (I-125) plaque brachytherapy utilizing COMS technique in a non-COMS institution. The purpose of this study was to assess visual outcomes and eye preservation following treatment of uveal melanoma with I-125 plaque brachytherapy and to identify factors predictive of these outcomes.

Materials/Methods: Retrospective analysis of 57 consecutive patients with uveal melanoma treated with I-125 plaque brachytherapy at Wake Forest University Baptist Medical Center between January 1990 and December 2003 was performed. Two endpoints were used to evaluate final visual acuity in the irradiated eye: 1) loss of visual acuity in Snellen lines and 2) visual acuity of 20/200 or worse. Median follow up was 3.65 years (range 0.8–14 years). Kaplan-Meier estimates were used to determine the enucleation-free survival rate. Factors examined included patient, tumor and treatment characteristics. Multivariable Cox proportional hazards regression was performed using stepwise selection of significant univariate correlates to determine predictors of visual outcome.

Results: Pre-treatment median visual acuity was 20/30 (range 20/200 to 20/15). Median initial tumor thickness was 4.2mm (1.9–10.5) and longest tumor diameter was 11mm (6–17). Tumor location was posterior to the equator in 45 patients (79%). At last follow up, visual acuity was stable (defined as loss of less than 2 Snellen lines) in 27 (47%) treated eyes. Final visual acuity was better than 20/200 in 33 (58%) patients and better than or equal to 20/40 in 23 (40%) treated eyes. The final median visual acuity was 20/60 (no light perception to 20/20). Univariate analysis demonstrated tumor thickness (p = 0.005), longest tumor diameter (p = 0.021), presence of a relative afferent pupillary defect (RAPD) (p = 0.037), and macular degeneration (p < 0.0001) as predictors for inferior visual outcome. Multivariate analysis demonstrated tumor thickness (p = 0.012), RAPD (p < 0.041), and macular degeneration (p < 0.0001) as independent predictors of inferior visual outcome. A total of eight (14%) patients underwent enucleation; 5 for local failure and 3 for treatment related complications. Enucleation-free survival at five years was 84.5% (95% CI 74.3–94.6%). The 5 year local control for all patients was 90.3% (95% CI 82.2–98.5%).

Conclusions: At this institution the majority of treated patients were spared from severe visual loss and a large percentage maintained their pretreatment visual acuity. In addition, our rate of enucleation compared favorably with that reported at other institutions and in the COMS trial. Patients with good pretreatment vision and thinner tumors have excellent visual outcome with good local control following plaque radiotherapy for choroidal melanoma.

1025 Repeated Proton Beam Therapy for Hepatocellular Carcinoma

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Purpose/Objective: To retrospectively evaluate the safety and effectiveness of repeated proton beam therapy for newly developed or recurrent hepatocellular carcinoma (HCC).

Materials/Methods: From June 1989 through July 2000, 225 patients with HCC underwent their first course of proton beam therapy at the University of Tsukuba. Of them, 27 with 68 lesions who had undergone 2 or more courses of proton beam therapy were enrolled in this study. A median total dose of 72 Gy in 16 fractions over 22 days and 66 Gy in 16 fractions over 22 days were given with proton beams for the first course and the rest of the courses, respectively.

Results: The 5-year survival rate and the median survival period from the beginning of the first course of proton beam therapy for the 27 patients were 55.6% and 62.2 months, respectively, and the five-year local control rate for the 68 lesions was 87.8%. The five-year local control rates for the first and second courses of proton beam therapy were 88.3 and 86.3%, respectively, with no statistically significant difference. Of the patients, 2 suffered from acute hepatic failure, 1 from a rib fracture, and 1 each